MANCHESTER 1824

The University of Manchester

EART11200 - The Natural Scientist's Toolkit

Mathematics and Physics

Week 7 Practical - Integration using Newton's Laws (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

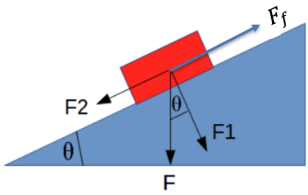
(1) Understand and describe Newton's Laws of Motion to explain the relationship between force, mass, and acceleration in physical systems.

(2) Apply Newton's Second Law to formulate ordinary differential equations (ODEs) that model the motion of objects in various contexts.

(3) Use integration to solve ODEs and determine physical behaviour in real-world problems.

Exercise 1: Object on an Incline With Friction

- In lectures, we considered an object resting on an incline.
- The forces can be resolved into the following components:
  - $F = mg$
  - $F1 = mg\cos(\theta)$
  - $F2 = mg\sin(\theta)$
- The initial velocity of the object is 0m/s and the angle is $\theta = 30$ degrees.
- However, this time, there is a constant frictional force ($F_f = 0.1mg$)
- Calculate the velocity of the object after 10 seconds.



Solution 1: Object on an Incline With Friction

- First we sum all of the forces in the direction of travel

$$F = \sum_i F_i = mg\sin(\theta) - 0.1mg$$

- This means the force $F2$ can be used to formulate our ODE:

$$m\frac{dv}{dt} = mg\sin(\theta) - 0.1mg$$

- Canceling the mass (m) gives:

$$\frac{dv}{dt} = g\sin(\theta) - 0.1g$$

Solution 1: Object on an Incline With Friction

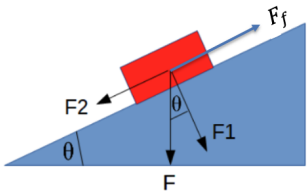
- Using separation of variables, we can integrate and calculate the velocity as a function of time.
- We integrate to find the velocity (v) at after 10 seconds ($t = 10$) assuming the object starts from rest ($v = 0$, $t = 0$):

$$\int_0^v dv = \int_0^{10} (g\sin(\theta) - 0.1g)\, dx \quad \Longrightarrow \quad [v]_0^v = (g\sin(\theta) - 0.1g)\,[x]_0^{10}$$
$$\Longrightarrow \quad v = 10g\sin(\theta) - g = 39.2\text{m/s}$$

- So the object was travelling at 39.2m/s after 10 seconds

Exercise 2: Object on an Incline With Friction

- In lectures, we considered an object resting on an incline.
- The forces can be resolved into the following components:
  - $F = mg$
  - $F1 = mg\cos(\theta)$
  - $F2 = mg\sin(\theta)$
- The initial velocity of the object is 0m/s and the angle is $\theta = 30$ degrees.
- However, this time, there is a constant frictional force ($F_f = 0.1mg$)
- Calculate the velocity **after the object travels 10 metres**.
- HINT: Find a formulation of Newton's second law that relates distance (x) and velocity (v).



Solution 2: Object on an Incline With Friction

- First we sum all of the forces in the direction of travel

$$F = \sum_i F_i = mg\sin(\theta) - 0.1mg$$

- This means the force $F2$ can be used to formulate our ODE:

$$mv\frac{dv}{dx} = mg\sin(\theta) - 0.1mg$$

- Canceling the mass (m) gives:

$$v\frac{dv}{dx} = g\sin(\theta) - 0.1g$$

Solution 2: Object on an Incline With Friction

- Using separation of variables, we can integrate and calculate the velocity as a function of time.
- We integrate to find the velocity (v) at after travelling 10m ($x = 10$) assuming the object starts from rest ($v = 0, x = 0$):

$$\begin{aligned}
\int_0^v v\,dv = \int_0^{10} (g\sin(\theta) - 0.1g)\,dx \quad &\implies \quad \left[\frac{v^2}{2}\right]_0^v = (g\sin(\theta) - 0.1g)\,[x]_0^{10} \\
&\implies \quad \frac{v^2}{2} = 10g\sin(\theta) - g \\
&\implies \quad v = \sqrt{20g\sin(\theta) - 2g} = 8.85\text{m/s}
\end{aligned}$$

- So the object was travelling at 8.5m/s when it reached 10m.

Intended Learning Outcomes

(1) Understand and describe Newton's Laws of Motion to explain the relationship between force, mass, and acceleration in physical systems.

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(3) Use integration to solve ODEs and determine physical behaviour in real-world problems.